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Q.No.1

A bank wants to calculate simple interest for customers. Write a C program that takes principle, rate , and time as input and calculates the interest using a function.

Solution :

```
#include <stdio.h>

// User-defined function

float calculateInterest(float p, float r, float t) {

   return (p * r * t) / 100;

}

int main() {

   float principal, rate, time;

    printf("Enter Principal: ");

   scanf("%f", &principal);

   printf("Enter Rate: ");

   scanf("%f", &rate);

   printf("Enter Time: ");

   scanf("%f", &time);

  float interest = calculateInterest(principal, rate, time);

 printf("Simple Interest = %.2f\n", interest);

 return 0;

}
```

INPUT :

Enter principle : 5000

Enter rate : 7.5

Enter time : 3

OUTPUT :

Simple interest = 1125.00

In banking systems , interest calculation is common operation. This program simulates a basic banking feature where the user enters the principle amount , rate of interest and time period.

Q.No.2

A student management system needs to calculate total and average marks of 5 subjects . Write a C program using function to compute total and percentage .

Solution :

```
#include <stdio.h>

float calculateTotal(float marks[]) {

   float sum = 0;

   for(int i = 0; i < 5; i++) sum += marks[i];

   return sum;

}

float calculatePercentage(float total) {

   return total / 5;

}

int main() {

   float marks[5];

   printf("Marks: ");

   for(int i = 0; i < 5; i++) {

      scanf("%f", &marks[i]);

   }

   float total = calculateTotal(marks);

   float percentage = calculatePercentage(total);

   printf("Total = %.0f\n", total);

   printf("Percentage = %.2f\n", percentage);
```

```
    return 0;
}
```

INTPUT :

Marks : 65 82 77 91 88

OUTPUT :

Total =403

Percentage = 80.60

The system takes marks of 5 subject as input and calculate total and percentage using functions.

Q.No.3

A teacher wants to identify whether a given number is a prime number . Write a C program that takes a number as input and checks whether it is prime using a function .

Solution :

```
#include <stdio.h>

int is Prime(int num) {

   if (num <= 1) return 0;

for (int i = 2; i * i <= num; i++) {

      if (num % i == 0) return 0;

   }

   return 1;

}

int main() {

   int number;

   printf("Enter an integer number: ");

   scanf("%d", &number);

    if (isPrime(number)) {

   printf("Prime\n");

   } else {

      printf("Not Prime\n");

   }

 return 0;

}
```

INTPUT (not prime):

Enter an integer number : 15

OUTPUT :

Not prime

INPUT(Prime):

Enter an integer number : 13

OUTPUT :

Prime

A prime number is a number that is divisible only by 1 and itself .

Q.No.4

A library system needs to calculate late fine for books returned after the due date . Write a C program that takes the number of late days as input and calculates the fine using function.

Solution :

```
#include <stdio.h>

int calculateFine(int days) {

   if (days > 5) {

      return days * 2;

   }

   return 0;

}

int main() {

   int days;

   printf("Enter number of late days: ");

   scanf("%d", &days);

   int fine = calculateFine(days);

  if (fine > 0) {

      printf("Fine = %d\n", fine);

   } else {

      printf("No Fine\n");

   }

  return 0;

}
```

INPUT (No fine case) :

Enter the number of days : 4

OUTPUT :

No Fine

INPUT(Fine case) :

Enter the number of days : 10

OUTPUT :

Fine = 20

In a library students must return books on time . if a book is returned late , a fine is charged based on the number of delayed days .

Q.No.5

An ATM machine needs to check whether the entered PIN is correct . Write a C program that takes a PIN from the user and compares it with a predefined PIN using a function .

Solution :

```
#include <stdio.h>

int checkPIN(int input) {

   int secret = 1234;

   return (input == secret); // Returns 1 if match, 0 if not

}

int main() {

   int pin;

   printf("Enter PIN: ");

   scanf("%d", &pin);

   if (checkPIN(pin)) {

      printf("Access Granted\n");

   } else {

      printf("Invalid PIN\n");

   }

   return 0;

}
```

INPUT :

Enter PIN : 9999

OUTPUT :

Invalid PIN .